TABLE I
Systematic Comparison of Reasoning and Self-Correction Evaluation Frameworks

Framework	Primary Target	Intervention Type	Prefix Symmetry	Core Empirical Finding
GSM8K [1]	End-to-end reasoning	None (Question only)	None	Scaling improves final outcome accuracy
PRM800K [3]	Process supervision	Step classification	Unmatched	Step-level oversight outperforms outcome oversight
Reflexion [5]	Multi-turn correction	Verbal reflection buffer	Asymmetric	External verbal feedback enables error recovery
CRITIC [6]	Tool-based correction	External interpreter/search	Interactive	Tool feedback enables calculation self-correction
Huang et al. [7]	Intrinsic self-correction	Prompted reflection	Post-hoc	Prompted reflection without feedback degrades accuracy
BIG-Bench Mistake [8]	Error localization	Explicit error location cue	Post-hoc traces	Models cannot find errors, but fix given location
ProcessBench [9]	Error step identification	Benchmark critic tasks	Varied	PRMs struggle to locate errors on competition math
RecoveryEval (Ours)	In-flight continuation inertia	Symmetrical step perturbation	Length-matched ($\Delta_{\text{tokens}} = 0$)	98.5% error propagation; 0% spontaneous recovery

$0.7, p = 0.9$) across 5 fixed seeds $(401, 402, 403, 404, 405)$ with a 256-token budget.

D. Cryptographic Provenance and Artifact Integrity

Every rollout record captures exact input token IDs, generated BPE token IDs, wall-clock inference duration, and model weight digest in an append-only event ledger. The canonical dataset 10_COMPLETE_QWEN_PROTOCOL2_ROLLOUTS.jsonl is frozen under SHA-256 digest b8792f8df375fccdbd800d440cc884ef2db2f860ff3533af01c70161a5cc1d15.

V. Statistical Methodology and Results

A. Cluster-Robust Factorial Outcomes

Table II summarizes completion accuracies across all four experimental cells under heuristic regex and structural semantic verification.

TABLE II
Protocol 2.0 Factorial Outcomes ($N = 400$ Rollouts, $C = 20$ Clusters)

State Condition	Base (π_0)	Instruct (π)	Difference (Δ)
Recovery States (s_R)	0.0000	0.0000	+0.0000
Control States (Heuristic)	0.9700	1.0000	+0.0300
Control States (Structural)	1.0000	1.0000	+0.0000
Matched Contrast (Heuristic)	–	–	−0.0300
Cluster SE / p-value	–	–	$SE = 0.0164, p = 0.0828$
95% Confidence Interval	–	–	$[-0.0643, +0.0043]$
Matched Contrast (Structural)	–	–	+0.0000 (Sample Null)
Cluster SE / p-value	–	–	Degenerate ($s_D = 0.0000$)

B. Degenerate Variance Analysis

Under structural scoring, every problem cluster $j \in \{1, \ldots, 20\}$ yields:

$$\Delta_{j,\text{Base}} = -1.0000, \quad \Delta_{j,\text{Instruct}} = -1.0000 \implies D_j = 0.0000$$

Consequently, the sample cluster standard deviation is $s_D = 0.0000$, and the cluster-robust standard error is $\text{SE} = 0.0000$. Under this condition:

- The Student-t statistic $t = \bar{D}/\text{SE} = 0/0$ is mathematically undefined.
- The conventional two-sided p-value is not computable.
- The conventional Studentized 95% confidence interval $[0, 0]$ is degenerate.

We emphasize that zero sample variance across 20 items is an empirical consequence of boundary saturation (100% control vs. 0% recovery), rather than proof of universal population equivalence.

C. Verifier Forensics and Heuristic Regex Failures

In heuristic scoring, Base Control scored 97.0% due to 3 apparent failures under seed 405 (problems fresh_004, fresh_006, fresh_013). Trace-by-trace inspection revealed that the base model correctly calculated the revenue but concluded with an unprompted summary review ending in list bullets (e.g., "- In step 3..."). The fallback regex matched the trailing bullet index 3 instead of the revenue answer. Under structural equivalence, all three rollouts are mathematically correct, establishing 100/100 (100.0%) accuracy for both checkpoints.

D. Behavioral Trace Decomposition: Continuation Inertia

We decompose all 200 recovery rollouts across five explicit process dimensions:

- Final Accuracy (Y_{acc}): Exactly 0 / 200 (0.0%) reached the gold terminal answer.
- Explicit Error Detection (Y_{det}): Exactly 0 / 200 (0.0%) emitted backtracking cues (*wait, actually, mistake, wrong*).
- Intermediate Correction (Y_{corr}): Exactly 0 / 200 (0.0%) corrected the corrupted Step 1 product.